

Document	Detail
Course	GIS Applications in Telecommunications
Day	Day 4 — Fiber, Transmission & Microwave Link Planning
Version	3.0
Depends on	Day 1 (GDB design) + Day 2 (sites) + Day 3 (priority expansion zones)
Primary tools	Geometric Network / Utility Network Analyst (ArcMap), Feature-to-Network connectivity design, 3D Analyst Line of Sight, Profile Graph / Stack Profile, Viewshed / Geodesic Viewshed, Spatial Analyst
Data folder	Day4_Fiber_Microwave/data/

1. Learning Objectives

By the end of Day 4, participants will be able to:

- Design FTTH/transmission node-edge topology (OLT → Joint → FDH → Splitter → ONT) in a geodatabase.
- Build and use a geometric network (ArcMap Utility Network Analyst) — or Pro connectivity workflow — to trace fiber connectivity.
- Map microwave hubs/remotes and evaluate Line of Sight (LOS) with 3D Analyst.
- Extract and interpret terrain profiles (Profile Graph / Stack Profile) for Fresnel/clearance discussion.
- Run Viewshed analysis for broader visibility/obstruction checks around microwave sites.
- Recommend accept / redesign / raise-antenna decisions for planned links serving Day 3 expansion zones.

2. Day Agenda

Time	Session	Outcome
09:00–09:30	Recap Day 3 + transport concepts	Fiber vs microwave roles clear
09:30–10:45	Lab 1 — Nodes, fiber routes, GDB topology	FTTH/transmission layers ready

10:45–11:00	Break	—
11:00–12:15	Lab 2 — Geometric network & tracing	Connectivity / isolation traces work
12:15–13:15	Lunch	—
13:15–14:30	Lab 3 — Microwave LOS + profile graph	LOS pass/fail + profile interpreted
14:30–14:45	Break	—
14:45–15:40	Lab 4 — Viewshed & expansion decision	Visibility map + recommendations
15:40–16:00	Outputs, quiz, bridge to Day 5	Day 4 deliverables complete

3. Concepts Quick Brief

3.1 Why GIS for fiber & microwave?

- Fiber is a connected network problem (what feeds what; where cuts isolate customers).
- Microwave is a terrain/obstacle visibility problem (LOS, Fresnel, viewshed).
- Both must align with commercial demand zones from Day 3.

3.2 Key terms

Term	Meaning in this course
OLT	Optical Line Terminal — central GPON headend
FDH	Fiber Distribution Hub / cabinet
Splitter	Passive optical split (e.g., 1:8, 1:16)
ONT	Optical Network Terminal at customer/building
Feeder / Distribution	Fiber hierarchy: core feeders vs access distribution
Geometric Network	ArcMap network of junctions + edges with connectivity rules
Utility Network Analyst	ArcMap toolbar for trace, flow, flags on geometric networks
LOS	Line of Sight between TX and RX antennas
Profile Graph	Cross-section of elevation along a link
Viewshed	Cells visible from observer point(s)
Fresnel zone	Ellipsoid clearance volume around MW path (discussed conceptually)

4. Datasets — Synthetic (lab) + Real (optional)

4.1 Synthetic data included

File	Type	Content	Use
NetworkNodes.csv	Points table	OLT, FDH, Splitter, Joint, MW Hub/Remote	Network junctions
FiberRoutes.geojson	Lines	Feeder + distribution fiber (Existing/Planned)	Edges for fiber network
MicrowaveLinks.geojson	Lines	Existing / Planned / Candidate MW links	LOS & MW topology
FTTH_ONT_Buildings.csv	Points	80 sample buildings/ONTs	Access endpoints
ObstructionPolygons.geojson	Polygons	Synthetic buildings/hill proxies with HEIGHT_M	LOS obstruction demo
SyntheticDEM_Points.csv	XYZ points	~1,100 elevation samples	Build TIN/raster DEM offline
ViewshedObservers.csv	Points	MW observers + OFFSETA heights	Viewshed inputs
ConnectivityRules.csv	Table	Allowed node-to-node rules	Geometric network design
LOS_Workflow_Checklist.csv	Table	6-step LOS procedure	Lab 3 guide
Day4_AllTables.xlsx	Excel	All tabular sheets	Instructor view
REAL_DATA_SOURCES.txt	Text	SRTM/OSM/HDX links	Optional real path

4.2 Optional REAL data

- DEM: SRTM 30 m / NASADEM / Copernicus GLO-30 / AW3D30 (preferred for LOS/viewshed).
- Buildings: OSM footprints (Geofabrik); extrude if heights available.
- Roads: OSM highways for trench routing context.
- Day 3 MarketingZones / suitability ranks for “where to extend fiber/MW”.

NOTE: Synthetic DEM is for classroom continuity. For realistic microwave training, switch to real SRTM/NASADEM as soon as possible.

5. Software & Project Setup

5.1 Required extensions

- ArcGIS Pro: 3D Analyst (LOS, Profile, Viewshed); Spatial Analyst helpful.
- ArcMap (Lab 2 classic path): Geometric Network + Utility Network Analyst toolbar.
- If ArcMap is unavailable: use Pro “functional connectivity” path in Lab 2B (attribute-based trace + topology).

5.2 Create Day 4 project

- Open ArcGIS Pro → New Map project MetroTel_Day4_FiberMW.aprx
- Folder connection to .../Day4_Fiber_Microwave/data
- Create file GDB MetroTel_Day4.gdb
- Optional: also open ArcMap 10.x mxd MetroTel_Day4_GeometricNetwork.mxd for Lab 2A
- Set a projected CRS in meters (local UTM) for DEM/LOS work; keep WGS84 sources and Project them.

5.3 Load & convert baseline layers

- XY Table To Point: NetworkNodes.csv → NetworkNodes (LON/LAT, GCS WGS84) → Project to UTM → NetworkNodes_UTM
- Import FiberRoutes.geojson → FiberRoutes → Project to FiberRoutes_UTM
- Import MicrowaveLinks.geojson → MicrowaveLinks → Project to MicrowaveLinks_UTM
- XY Table To Point: FTTH_ONT_Buildings.csv → ONT_Buildings
- Import ObstructionPolygons.geojson → Obstructions
- XY Table To Point: ViewshedObservers.csv → ViewshedObservers
- Symbolize nodes uniquely by NODE_TYPE; lines by STATUS (Existing/Planned/Candidate)

CHECKPOINT: Map shows OLTs/FDHs/joints, fiber feeders, MW links, ONTs, and obstruction polygons.

6. Lab 1 — Build Fiber/Transmission Feature Dataset

Objective: Organize hierarchical FTTH/transmission layers ready for network connectivity.

- In MetroTel_Day4.gdb create Feature Dataset FD_Transport (set UTM spatial reference).
- Move/copy NetworkNodes_UTM, FiberRoutes_UTM, ONT_Buildings (projected) into FD_Transport.
- Add fields if missing: ENABLED (short, default 1), FROM_NODE, TO_NODE already on edges.
- Create topology FD_Transport_Topo (optional but recommended):
- FiberRoutes must not have dangles (except intentional ends).
- Fiber endpoints must be covered by NetworkNodes.
- Nodes must be covered by FiberRoutes endpoints (for network junctions).
- Validate topology; fix obvious digitizing gaps by snapping planned routes to JH-04 / FDH-04 / FDH-06.
- Attribute QA: select STATUS = 'Planned' fiber — confirm they serve Day 3 west/tech-park expansion story.
- Create relationship-friendly IDs: ensure every FiberRoutes.FROM_NODE/TO_NODE exists in NetworkNodes.NODE_ID.

CHECKPOINT: Feature dataset + clean node/edge IDs; planned west fiber visible.

7. Lab 2 — Geometric Network & Connectivity Tracing

Objective: Establish and test connectivity between FTTH/transmission elements.

7.1 Lab 2A — ArcMap Geometric Network (curriculum path)

- Open ArcMap → add FD_Transport layers (File GDB).
- In Catalog, right-click FD_Transport → New → Geometric Network
- Name: GN_FiberTransport
- Select participating classes: NetworkNodes (junction), FiberRoutes (simple edge)
- Snap sources/sinks optionally: set OLT nodes as Sources (for flow-style traces)
- Use ConnectivityRules.csv to set weights/roles conceptually; configure connection rules so OLT–Joint–FDH–Splitter hierarchy is allowed
- Build geometric network; add Utility Network Analyst toolbar
- Set Analysis flags: Add Junction Flag on OLT-01
- Add Edge Barrier on one feeder (e.g., FX-05) to simulate cut
- Trace Tasks → Trace Downstream / Find Connected / Find Disconnected
- Record which FDHs/ONTs become isolated when FX-07/FX-08 path is barred (south corridor)
- Clear barriers; place flag on FDH-03; Find Path / Find Upstream back to OLT-01

TIP: If Utility Network Analyst is missing, customize toolbars → enable “Utility Network Analyst”.

7.2 Lab 2B — ArcGIS Pro functional connectivity path (if no ArcMap)

- Use Select By Attributes / Select By Location to emulate traces:
- Select FiberRoutes where FROM_NODE='OLT-01' OR TO_NODE='OLT-01' (first hop).
- Iteratively select nodes touching selected edges, then edges touching those nodes (Expand selection).
- Build a “TraceLog” table manually for classroom: hop number, NODE_ID, ROUTE_ID.
- Simulate outage: remove/disable FX-03 and re-run selection expansion from OLT-01 — note OLT-02/FDH-02 isolation risk if no alternate path.
- Optional Pro: create Trace Network (if license/version supports) on FD_Transport with same junction/edge roles.

NOTE: Curriculum emphasizes ArcMap geometric network; Pro Trace Network is the modern equivalent where available.

CHECKPOINT: You can demonstrate at least one connected trace and one isolation/outage scenario.

8. Lab 3 — Microwave LOS & Terrain Profile

Objective: Evaluate microwave links with Line of Sight and profile tools.

8.1 Build DEM

- Path A (offline): Create TIN or raster from SyntheticDEM_Points.csv

- XY Table To Point → DEM_Points; Project to UTM
- Create TIN (3D Analyst) using ELEV_M, or Topo to Raster / IDW → DEM_Synth
- Path B (real data): Load SRTM/NASADEM clipped to Capital Region → DEM_Real
- Set DEM as ground surface in scene or keep as raster for geoprocessing

8.2 Line of Sight (existing vs planned)

- Select MicrowaveLinks where LINK_ID in ('MWL-01','MWL-05','MWL-07') — one good, one planned over hills, one candidate
- 3D Analyst Tools → Visibility → Line Of Sight
- Input Surface = DEM_Synth (or DEM_Real)
- Target / Observer features: use MicrowaveLinks as 3D lines OR run LOS between NetworkNodes MW points
- Observer offset = TX_HEIGHT_M / OFFSETA; Target offset = RX_HEIGHT_M / OFFSETB (from link attributes / ViewshedObservers)
- Output = LOS_Results with Visible / Not Visible segments
- Symbolize visible = green, obstructed = red
- Overlay Obstructions polygons; discuss whether buildings/hills cause failures on MWL-05 (Central→West Planned)

8.3 Profile Graph

- Convert/ensure MW link lines are 3D using Interpolate Shape against DEM → MWLinks_3D
- In Pro: Analysis chart / Stack Profile OR 3D Explorer profile; in ArcMap: Profile Graph tool
- Create profiles for MWL-01 (expect clearer) and MWL-05 (expect terrain bulge near west hill)
- Mark approximate first Fresnel clearance conceptually (0.6 Fresnel discussed; not full RF engineering)
- Decision log in a table Link_Decision.csv fields: LINK_ID, LOS_STATUS, OBSTRUCTION_NOTE, ACTION (Accept / Raise antenna / Reroute / Prefer fiber)

CHECKPOINT: LOS colors show pass/fail; profile exported/screenshot for MWL-05; decision noted.

9. Lab 4 — Viewshed & Expansion Recommendation

Objective: Broader visibility/obstruction check and final transport recommendation.

- Viewshed / Geodesic Viewshed tool
- Input raster = DEM; Observer features = ViewshedObservers (or MW hubs only)
- Surface/Observer offset: use OFFSETA_M (antenna heights)
- Optional: outer radius 8–15 km; inner radius 0
- Output = Viewshed_MW (binary or frequency if multiple observers)

- Zonal comparison: which Day 3 high-suitability / planned FDH areas fall inside MW-01 viewshed?
- Combine results:
- If west expansion (FDH-04 / MW-05) fails LOS and viewshed is weak → prefer planned fiber FX-09/FX-10
- If airport/north remains visible → keep MW for rapid backhaul
- Candidate MWL-07: accept only if LOS clear; else drop
- Layout 1: Fiber geometric network + OLTs/FDHs + outage example
- Layout 2: MW links with LOS green/red + DEM hillshade
- Layout 3: Viewshed + planned expansion nodes from Day 3/4
- Export Day4_FiberMW_Layouts.pdf and short recommendation paragraph

CHECKPOINT: Viewshed created; written recommendation for west expansion transport mode.

10. Expected Outputs Checklist

#	Deliverable	Done?
1	MetroTel_Day4_FiberMW.aprx + MetroTel_Day4.gdb	
2	FD_Transport with nodes + fiber routes (+ topology optional)	
3	Geometric network (ArcMap) OR documented Pro connectivity trace	
4	At least one outage/isolation trace result	
5	DEM (synthetic or real) used in analysis	
6	LOS results for ≥ 2 microwave links	
7	Profile graph/screenshot for obstructed/planned link	
8	Viewshed from MW observer(s)	
9	Recommendation: fiber vs MW for west/tech-park expansion	
10	Exported layout PDF	

11. Instructor Answer Key — Typical Findings

- Existing MW hub links (Central–North/East/South) often pass or mostly pass on synthetic DEM.
- Planned Central–West (MWL-05) is designed to interact with the west hill/obstructions — expect partial/fail LOS → teach redesign.
- Fiber planned branches FX-09/FX-10/FX-11 are the fallback to serve FDH-04 and FDH-06.
- Cutting a south feeder can isolate FDH-05 / SPL-03 in simple tree topology — good outage demo.

- Viewshed from MW-01 should cover central/north better than behind western ridge.

12. Troubleshooting

Problem	Likely cause	Fix
Geometric Network wizard missing	Using Pro only / wrong GDB type	Use ArcMap on File GDB feature dataset; or Lab 2B
LOS all not visible	Offsets=0 or DEM units mismatch	Set antenna offsets; DEM in meters; projected CRS
Profile flat/wrong	Line not interpolated to surface	Run Interpolate Shape first
Viewshed empty	Observers outside DEM / wrong offsets	Clip DEM larger; check OFFSETA_M
Nodes not connecting	Gaps / units / no snap	Snap fiber ends to nodes; rebuild network
Cannot find Utility Network Analyst	Toolbar not enabled	ArcMap Customize → Toolbars

13. Quick Quiz

- Name the typical FTTH hierarchy from core to customer.
- What ArcMap toolbar traces a geometric network?
- What is the difference between Line of Sight and Viewshed?
- Why might a planned MW link fail even if endpoints are “close” on a 2D map?
- If LOS fails to a Day 3 high-value expansion zone, what alternative does Day 4 data provide?

Suggested answers:

- 1) OLT → (Joint) → FDH → Splitter → ONT/Building.
- 2) Utility Network Analyst.
- 3) LOS = visibility along one path between points; Viewshed = area visible from observer(s).
- 4) Terrain/buildings intrude into the path/Fresnel zone; antenna heights too low.
- 5) Planned fiber feeders (FX-09/10/11) to FDH-04/FDH-06.

14. Bridge to Day 5

Day 5 packages coverage (Day 2), geo-marketing (Day 3), and fiber/microwave (Day 4) into ArcGIS Online web maps, dashboards, and Story Maps for executive reporting. Keep today's LOS results, viewshed, and fiber trace maps ready to publish.

15. Version History

Version	Changes
v1	Draft Day 4 fiber/MW outline
v2	Added LOS/viewshed lab sequence
v3	Full manual with MetroTel synthetic fiber+MW datasets, DEM points, obstruction polygons, ArcMap geometric network steps, Pro fallback, LOS/profile/viewshed labs, checkpoints, quiz, real-data options